

VENABU, Norway

WMO: 013800

Lat: 61.651N

Lon: 10.109E

Elev: 3051

StdP: 13.15

Time Zone: 1.00 (EUC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.1	-3.8	-13.7	2.9	-7.1	-9.3	3.7	-3.1	25.0	24.6	22.1	24.3	3.1	0	0.757

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	71.6	55.4	67.9	54.1	63.9	52.4	57.7	67.1	55.9	63.7	54.1	61.4	6.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.2	70.0	59.4	52.4	65.4	58.2	50.7	61.4	56.4	26.2	68.0	24.9	63.5	23.8	61.7	63.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.4	17.4	14.8	DB	-13.5	75.4	3.9	4.1	-16.3	78.3	-18.6	80.7	-20.8	83.0	-23.6	86.0	(4)
(5)			WB	-14.0	59.5	3.8	2.1	-16.7	61.0	-19.0	62.2	-21.1	63.4	-23.9	65.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	33.0	15.2	17.0	22.6	30.9	40.3	48.0	53.5	49.9	43.1	32.6	23.1	18.4		
	DBStd	15.09	8.87	9.16	8.18	7.03	8.17	5.93	5.85	4.58	5.35	6.00	8.04	9.24		
	HDD50	6474	1078	923	848	572	319	105	28	58	215	539	808	981		
	HDD65	11688	1543	1343	1313	1022	766	509	359	467	658	1004	1258	1446		
	CDD50	263	0	0	0	0	19	46	136	56	6	0	0	0		
	CDD65	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDH74	34	0	0	0	0	1	3	30	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	6.2	5.9	6.0	7.1	6.4	6.9	7.3	6.2	5.5	6.2	6.0	5.4	5.9		
	PrecAvg	23.9	1.6	1.0	0.9	1.1	2.1	2.8	3.6	3.5	2.0	2.0	1.8	1.4		
(15) Precipitation	PrecMax	32.3	2.9	2.2	2.0	2.5	5.2	5.7	6.1	5.6	4.0	6.3	5.6	2.9		
	PrecMin	18.8	0.2	0.3	0.0	0.1	0.2	0.0	1.5	0.7	0.6	0.2	0.3	0.2		
	PrecStd	3.3	0.7	0.5	0.6	0.6	1.2	1.4	1.3	1.4	0.9	1.3	1.1	0.6		
	PrecStd	3.3	0.7	0.5	0.6	0.6	1.2	1.4	1.3	1.4	0.9	1.3	1.1	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.7	38.7	46.0	54.9	70.6	73.2	77.5	68.4	63.3	52.8	42.1	38.8		
		MCWB	33.4	33.4	37.6	43.1	52.4	53.9	58.7	55.9	51.8	48.7	40.0	35.3		
	2%	DB	33.0	35.4	41.3	49.9	65.8	68.9	73.9	65.1	58.9	47.5	38.4	36.0		
		MCWB	30.8	31.4	35.4	40.0	50.6	52.5	57.4	53.7	50.0	42.3	36.6	33.4		
	5%	DB	30.3	32.4	38.1	45.7	60.0	63.7	70.3	62.1	55.0	44.2	35.9	32.9		
		MCWB	29.1	29.8	33.1	37.7	47.7	50.4	55.9	52.7	48.6	40.5	34.6	31.4		
	10%	DB	27.4	29.1	34.2	42.1	54.6	59.6	65.9	58.8	51.9	41.5	33.2	30.4		
		MCWB	26.4	27.5	30.3	35.7	46.1	48.6	54.3	51.4	47.2	38.8	32.1	29.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.5	34.7	38.6	44.2	54.1	56.1	60.8	58.2	54.7	50.2	40.3	36.1		
		MCDB	35.6	37.9	43.9	53.7	65.6	65.2	72.8	64.9	57.7	51.1	41.7	38.0		
	2%	WB	31.3	32.0	35.9	40.6	51.8	54.2	58.8	56.2	51.8	44.0	37.1	33.8		
		MCDB	32.8	34.4	41.4	48.2	63.1	63.6	69.7	61.1	56.0	46.2	38.4	35.3		
	5%	WB	29.1	29.8	33.3	38.3	49.2	52.5	57.2	54.5	49.7	41.3	34.8	31.6		
		MCDB	30.4	32.2	37.2	44.6	58.2	61.6	67.6	59.3	53.5	43.4	35.8	32.9		
	10%	WB	26.4	27.6	31.0	36.3	46.2	50.4	55.5	52.5	47.6	39.0	32.3	29.3		
		MCDB	27.2	29.3	34.0	41.7	53.7	58.3	64.1	57.3	51.5	41.1	33.0	30.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	12.4	15.0	16.0	16.2	16.8	17.1	15.3	13.5	11.0	9.8	10.9		
		MCDBR	11.9	14.9	17.7	19.7	25.1	24.2	24.6	20.1	19.4	15.8	10.2	11.6		
		MCWBR	11.3	12.0	12.2	11.7	12.6	11.3	10.5	10.6	11.7	11.3	9.0	10.8		
	5% WB	MCDBR	11.8	13.5	16.1	17.9	22.3	20.4	21.3	16.5	16.1	12.6	9.5	11.6		
		MCWBR	11.1	11.1	11.4	10.8	12.0	10.3	10.0	9.7	10.6	10.2	8.6	11.0		
(40) Clear-Sky Solar Irradiance	taub		0.192	0.239	0.269	0.301	0.312	0.310	0.322	0.314	0.285	0.268	0.205	0.159		
	taud		2.124	2.277	2.273	2.287	2.453	2.524	2.493	2.507	2.575	2.499	2.182	1.974		
	Ebn at Noon		180	243	272	283	290	292	284	278	270	233	171	132		
	Edh at Noon		10	19	27	33	31	29	30	27	21	16	10	7		
(44) All-Sky Solar Radiation	RadAvg		89	302	795	1369	1553	1627	1492	1155	742	335	110	49		
	RadStd		14	56	77	127	144	153	174	97	82	50	20	4		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (56 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page